Career Launcher MBA

QA Arithmetic

Review Test – 01

A family of 4 people consumes certain quantity of rice. The price of rice increases by 12.5%. One day 2 guests come to visit the family. That day, the family consumed the same quantity of rice, while the guests consumed $\frac{2}{5}$th the amount of rice consumed by

the family. Find the percentage change in the family's expenditure.

(1) 37.5% increased

(2) 57.5% increased

(3) 37.5% decreased

(4) 35% decreased

Question – 1

In a city the consumption of rice is 7 times the consumption of wheat. If X% more rice and Y% more wheat is consumed, the aggregate amount consumed would be 9Z% more; but if Y% more rice and X% more wheat is consumed, the aggregate amount consumed would be 3Z% more. Find the ratio between X and Y.

(1) 5 : 1

(2) 4 : 1

(3) 7 : 1

(4) 1 : 3

Question – 2

Kapil saves a certain part of his income every year, so that he can purchase a house in 15 years. By what percent must he increase his savings, so that he can purchase the same house at same price 10 years later?

(1) 37.5% (2) 30% (3) 33.33% (4) 50%

Question – 3

A man wants to invest Rs.16850 splitting between his two sons aged 12 years and 16 years, in such a way that they would get equal amount when they are 80. If the compounded rate of return is $\frac{100}{3}$% per annum, then find the share of the younger son.

(1) 4050

(2) 12800

(3) 6050

(4) 14500

Question – 4

A shopkeeper wants to make some profit by selling rice. He contemplates various methods some of which are written below. Which of the following would maximize his profit?

i. Sell rice at a profit of 15.5%.

ii. Use a 900 gram weight instead of a 1 kg weight.

iii. Mix 10% impurities in rice and then sell it at a profit of 5%.

iv. Increase the price by 5% and reduce the weight by 5%.

(1) i or iii (2) ii (3) ii, iii and iv (4) Profits are same

Question – 5

If Asif offers a 40% discount on the marked price, then he makes a loss of 16.66%. What is the % age of impurities that has to be mixed to gain a 10% profit after the discount?

(1) 27% (2) 24.24% (3) 32% (4) 43.25%

Question – 6

In an election contested by 2 parties A and B, there were 2400 eligible voters. Party A was expected to win the election. However, on election day 33.33% of the people who wanted to vote for A were kidnapped. Party B was also able to influence some of the remaining Party A's voters and thus double the strength of its votes as compared to its initial strength of votes. Party A lost by a majority which was half of that by which it would have won, had the elections been fair. How many people finally voted for party A and party B respectively?

(1) 600-A, 1200-B (2) 300-A, 600-B (3) 450-A, 900-B (4) 600-A, 900-B

Question – 7

Rahul and Sonali each buy a goods for Rs. 1000 and Rs. 2000 respectively. Rahul marks up his goods by a% and offers no discount, while Sonali marks up her goods by 2a% and offers a discount of a%. If both make the same nonzero profit, find a.

(1) 12.5% (2) 20% (3) 25% (4) 30%

Question – 8

A merchant buys 40 articles, each at Rs. 20. He sells a of them at a profit of a% and remaining at a profit of (100 – a)%. What is the minimum profit the merchant could have made on this trade?

(1) 1025

(2) 295

(3) 310

(4) 975

Question – 9

Yogesh buys some items at Rs. 20 per item. The particular good is a part of an overall collection and the value is linked to the number of items that are already in the market. So the merchant sells the first item for Rs. 2, second for Rs. 4, third for Rs. 6 and so on. If he wants to make overall profit of at 40%, what is the minimum number of items he should sell?

(1) 39 (2) 32 (3) 18 (4) 27

Question – 10

Two cogged wheels working into each other have 48 cogs and 72 cogs. If the latter turns 120 times in $\frac{3}{4}$ths of a minute, how often does the former turn in 12 seconds?

(1) 24 (2) 135 (3) 48 (4) None of these

Question – 11

A, B and C get into a partnership for a year by investing Rs. 7200, Rs. 8800 and Rs. 5600 respectively. A is a working partner and gets $1/4^{th}$ of the total profit for his services and the remaining profit is divided among the three in the ratio of their investments. What is the amount of profit that B gets if A gets a total of Rs. 11988?

(1) Rs. 4992 (2) Rs. 9768 (3) Rs. 7326 (4) Rs. 11880

Question – 12

If a sum of money grows to $\frac{676}{441}$ times when invested for two years in a scheme where interest is compounded annually, how long will the same sum of money take to triple itself if invested at the same rate of interest in a scheme where the interest is computed using the simple interest method?

(1) 12 years

(2) 8.4 years

(3) 10.5 years

(4) 15 years

Question – 13

There are six different positive integers and their sum is 66. Out of these six numbers – one number is the average of all numbers, one number is the difference between the highest and the lowest numbers. What can be the maximum value of the highest number of this set?

(1) 20 (2) 22

(3) 25 (4) 26

Question – 14

The selling prices of two horses are equal. While one is sold at a 10% profit, the other is sold at a 20 % loss. What is the overall % profit or loss?

(1) 7.37% profit

(2) 7.37% loss

(3) 92.63% profit

(4) 92.63% loss

Question – 15

Let a, b, c, d be real numbers satisfying $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{a}$.

Find the value of $\frac{a+b+c+d}{a+b+c-d}$

(1) –1

(2) 0

(3) 2

(4) 0 or 2

Question – 16

A man is selling milk at a certain profit. He adds 20% of water to it and increases the selling price by 7.14% and as a result his profit increases by 100%. What was his usual profit %?

(1) 40%

(2) 55.55%

(3) 60%

(4) Cannot be determined

Question – 17

There are 8 positive integers. Each of the mean, median, mode and range of this set are 8. The largest integer in this set could be

(1) 12 (2) 14 (3) 15 (4) 16

Question – 18

In a faulty balance, scale with unequal pan weights, it requires 16 kg to balance an article, if article placed on the left pan. If article is placed on the right pan, it requires only 9 kg to balance it. What is the actual weight of the article?

(1) 7

(2) 12

(3) 11.5

(4) 12.5

Question – 19

In a certain class the number of girls is exactly between 60% and 65% of the total students of that class .What can be the minimum number of students in that class ?

(1) 7

(2) 8

(3) 11

(4) 12

Question – 20

A fruit seller purchased some apples and mangoes at the same price, and planned to sell mangoes at 20% profit and apples at 30% profit. He made certain baskets of apples and mangoes. Every basket had equal numbers of apples and Mangoes. He realized that from each basket he has a profit of 24%. He sold the remaining apples at a profit of 30%. On the whole he made a profit of 25%.What was the ratio of the number of apples to that of the number of mangoes which he purchased?

(1) 1 : 1 (2) 2 : 3 (3) 3 : 2 (4) 4 : 3

Question – 21

In a certain locality four-sevenths of total men are widowers and seven-elevenths of total women are widows. The number of widowers is twice that of number of widows. What fraction of the total population of the locality are not widows?

(1) $\frac{4}{11}$

(2) $\frac{6}{11}$

(3) $\frac{8}{71}$

(4) $\frac{57}{71}$

Question – 22

The ratio of the marked prices of A and B is 5 : 6. If A is sold at a x % discount and B is sold at a y% discount, their selling prices are in the ratio 18 : 21.6. If their cost prices are in the ratio 3:4, find the ratio of their profit %.

(1) 1 : 1

(2) 5 : 6

(3) 4 : 5

(4) Cannot be determined

Question – 23

In a class of 10 students, the average weight of the girls is 1 kg less than that of the boys while the weight of students are consecutive numbers from 51 to 60 kg. Find the number of boys in the class, if the average weight of boys and girls are integral values.

(1) 5 (2) 7 (3) 8 (4) 9

Question – 24

Samrat Ashok had ten chiefs, who each had to pay obligeance to him. They were numbered 1 to 10 and the chief with number n had to give him n gold coins each weighing 10 gms. The total gold weight of all coins received turned out to be 532 gms. Ashok's minister found that more than two chiefs had mixed brass with the gold and reduced the gold weight of each coin by a small amount, that is 20%. If the Chief numbered 1 was not involved in it, then what were the numbers of the Chiefs who gave the counterfeit coins? _____

Question – 25

Thanks!